Multiple Probabilities of Direction: A Rationale for Prior-odds Adjustment

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Abstract

In psychological research, Bayesian analyses are often summarized using posterior indices and then interpreted with conventional cutoffs across many parameters. The probability of direction (*pd*) is one such index, defined as the posterior probability that a parameter is greater or smaller than a null value. When *pd* is thresholded and used repeatedly, multiplicity becomes consequential but is rarely handled explicitly. This paper examines the operating characteristics of *pd*-based decisions under repeated use, leveraging its correspondence with the frequentist *p*-value in common settings. Simulations show that extreme *pd* values can occur by chance and that misleading conclusions become more likely as the number of tests increases. We then express *pd* in terms of posterior odds and propose a principled prior-odds adjustment that induces multiplicity control while preserving the underlying Bayes factor component. An accompanying R package implements the proposed prior-odds correction to support routine applied use.

Keywords: probability of direction, multiple testing, prior-odds

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Over the past two decades, the proportion of articles reporting Bayesian analyses in psychology has increased considerably (Jevremov & Pajić, 2024; Pfadt et al., 2025; Van De Schoot et al., 2017). Unfortunately, this expansion has been accompanied by practices reminiscent of the "null ritual" in frequentist statistics, where widespread use was associated with automatic application rather than principled reasoning (Cohen, 1994; Gigerenzer, 2004). For instance, researchers sometimes treat Bayes factor thresholds (e.g., BF > 3) as dichotomous decision rules or claim that an effect "exists" because the posterior probability of a positive effect exceeds a conventional benchmark such as 95% (Heck et al., 2023; Hoijtink et al., 2016; Kelter, 2023; Kruschke, 2018; Tendeiro et al., 2024; Tendeiro & Kiers, 2019). While such summaries are convenient, their use as default cutoffs encourages all-or-none interpretations. When these binary judgments are applied repeatedly across many parameters, the issue of multiplicity arises.

In the frequentist framework, multiplicity is typically framed as Type I error (false-positive) problem: if each individual test is conducted at level α, the probability of at least one false positive increases with the number of tests. Classical approaches therefore control either the family-wise error rate (the probability of at least one Type I error) or the false discovery rate (the expected proportion of false positives among rejected hypotheses) (Benjamini & Hochberg, 1995; Westfall & Young, 1993). Multiplicity concerns extend beyond false positives: when researchers examine many parameters and highlight the most extreme estimates, selection effects can produce exaggerated magnitudes (Type M error) (Altoè et al., 2020; Gelman & Tuerlinckx, 2000; Zwet & Cator, 2021). Bayesian shrinkage methods address these distortions by pulling extreme estimates toward more plausible values. Such shrinkage can arise through hierarchical models with partial pooling (Berry & Hochberg, 1999; Gelman et al., 2012, 2013; Kruschke, 2011) and/or through informative priors that reduce overestimation in low signal-to-noise settings (Dawid, 1994; Zwet & Gelman, 2022). Shrinkage reduces Type M errors, but it does not, on its own, provide a guarantee about the error properties of a repeated decision rule applied across many parameters. In particular, if the same cutoff is applied to many parameters (e.g., declaring

an effect whenever posterior probability exceeds 95%), the chance of at least one spurious finding can remain inflated, even if shrinkage reduces it. Bayesian approaches that directly target this issue typically rely on either simultaneous credible statements (Box & Tiao, 1992; Gelman & Tuerlinckx, 2000; Pruzek, 2016; Schnell et al., 2020) or priors on the hypothesis/model space that scale with the number of tested effects (Jeffreys, 1938; Scott & Berger, 2006; Scott & Berger, 2010; Westfall et al., 1997). Nevertheless, in psychological research shrinkage is often treated as sufficient for multiplicity control (Chen et al., 2020; Erčulj & Štrumbelj, 2015; Hogeveen et al., 2022; Limbachia et al., 2021; Mathar et al., 2022).

Other justifications invokes the subjectivist interpretation of Bayesian inference, which treats posterior probabilities as rational degrees of personal belief (Edwards et al., 1963; Lindley, 1972; Rozeboom, 1960). However, this interpretation fits poorly with common psychological practice, where researchers typically apply default software priors (often weakly informative) rather than priors encoding substantive domain knowledge (Smid & Winter, 2020; Van De Schoot et al., 2017). When these priors are combined with sufficient data, Bayesian and frequentist procedures often converge to similar conclusions (Albers et al., 2018; Bernardo & Smith, 1994; Gelman et al., 2013).

Given this convergence and the prevailing practices in psychological research, we aim to evaluate the frequentist operating characteristics of Bayesian decision rules in the context of multiple comparisons. Specifically, we focus on the Probability of Direction (*pd*), a Bayesian index that can be explicitly related to the frequentist *p*-value in common settings (Berger & Sellke, 1987; Casella & Berger, 1987; Makowski et al., 2019; Marsman & Wagenmakers, 2017). This link offers an interpretable framework for assessing *pd*-based decision rules under repeated use and for comparing Bayesian and frequentist approaches to multiplicity control.

The paper proceeds in four steps. First, we introduce *pd* and make explicit its connection to the frequentist *p*-value. Second, we derive the null behavior of *pd* and study the multiple-testing properties of *pd*-thresholded decisions in simulations, focusing on family-wise error rate. Third, we express *pd* in terms of posterior odds, relate these odds to directional Bayes

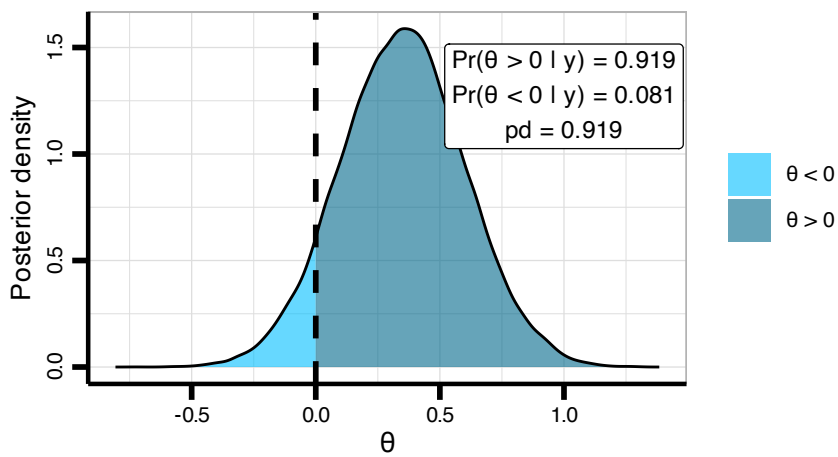
factors, and clarify the contribution of prior odds. Finally, building on Jeffreys' global-null formulation (Jeffreys, 1938; see also Westfall et al., 1997), we propose prior-odds adjustments for multiplicity, discuss practical implementation (including departures from independence), and provide guidance for reporting both raw and multiplicity-adjusted *pd* values.

Probability of direction

The probability of direction (*pd*) for continuous distributions can be defined as the maximum between the posterior probabilities that a parameter is greater or smaller than the null value (Figure 1): $pd = \max(\Pr(\theta < \theta_{null} \mid y),\ \Pr(\theta > \theta_{null} \mid y))$.

Figure 1

Posterior distribution with directional posterior mass on either side of θ_{null}, here defined as $\theta_{null} = 0$, shown in different colors; the probability of direction (pd) is the larger of the two posterior probabilities.



As proposed and popularized in applied Bayesian workflows [**REFs**], *pd* offers an intuitive summary of directional evidence by quantifying the proportion of posterior that lies on the dominant side (Makowski et al., 2019); it ranges from 0.50 (no directional preference) to 1 (all on one side). Because *pd* is a functional of the posterior, it inherits dependence on the full model specification (likelihood and prior), so the long-run error behavior of *pd*-based decisions is not a property of *pd* "in isolation" but of the overall Bayesian procedure that generates the posterior.

Probability of direction as a test of existence and direction

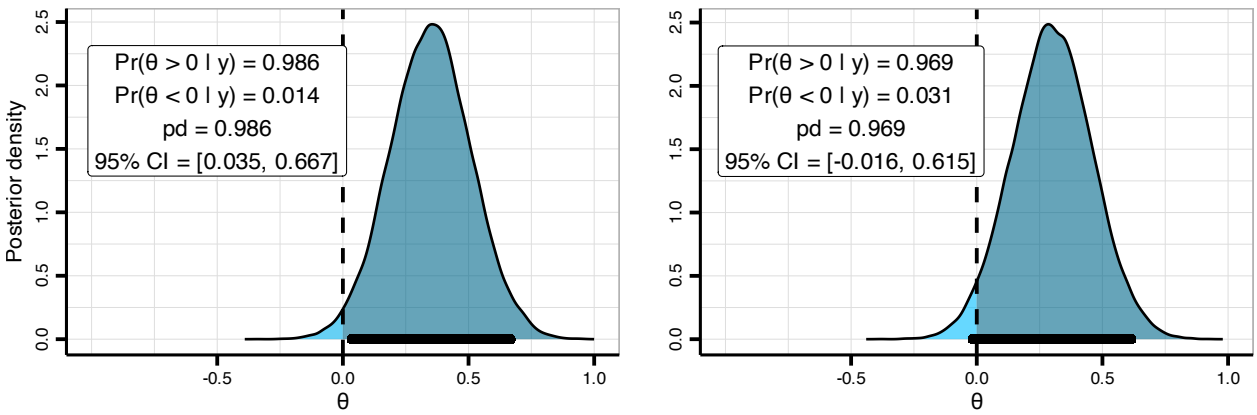
A convenient starting point for calibration is the mapping between pd and the frequentist p-value. Under flat (or sufficiently diffuse symmetric) priors in typical hypothesis-testing scenarios (e.g., testing means), pd is simply the posterior on the dominant side of θ_{null}. This value corresponds to the complement of the one-sided p-value (Casella & Berger, 1987; Marsman & Wagenmakers, 2017), with the two-sided relation approximated by $p_{two-sided} \approx 2p = 2(1 - pd)$. The pd is therefore often used to summarize both the existence and direction of an effect [**REFs**].

Setting a threshold $pd > c$ means that the posterior probability of the parameter having a given sign exceeds c. The value of c can be chosen so that the interval between the $(1 - c)$ and c posterior quantiles matches the equal-tailed central credible interval with nominal level $1 - \alpha$, whose endpoints are the $\alpha/2$ and $1 - \alpha/2$ quantiles. Equating these endpoints gives $1 - c = \alpha/2$ and $c = 1 - \alpha/2$; for instance, with $\alpha = 0.05$, this corresponds to $c = 0.975$ and $1 - c = 0.025$.

To illustrate, suppose $c = 0.975$ and $pd > c$ supports the claim that $\theta > 0$. In this case, the posterior probability that θ is nonpositive satisfies $\Pr(\theta \leq 0 \mid y) < 1 - c = 0.025$, meaning that less than 2.5% of the posterior mass lies at or below zero. Thus, the threshold $pd > c$ does more than indicate that positive values are more probable than negative ones, it also ensures that the posterior probability of the opposite sign is sufficiently small. In other words, if $\Pr(\theta \leq 0 \mid y) < 1 - c$, then zero lies below the $(1 - c)$-quantile of the posterior distribution, and the central credible interval delimited by the $(1 - c)$ and c quantiles excludes zero. Therefore, $pd > c$ can be interpreted as evidence that zero is excluded from the $(1 - \alpha)\%$ central credible interval (or equivalently $(c - \alpha/2)\%$, Figure 2).

Figure 2

Posterior distributions illustrating the relationship between the probability of direction (pd) and credible-interval (CI), using $c = 0.975$ and $\theta_{null} = 0$.



Although *pd* is a measure of direction, pure directional testing conditions on the assumption that $\theta \neq 0$ and evaluates only the sign of the effect by comparing H_+ versus H_-; $H_0 : \theta = 0$ is not among the hypotheses under consideration. In contrast, here we explicitly include H_0, focusing on decisions about both existence and direction (when the interval excludes zero, e.g., H_+) rather than decisions about direction only (when the interval includes zero, e.g., H_{0+}; see Goeman et al. (2010)).

Importantly, although we set $H_0 : \theta = \theta_{null} = 0$ throughout the manuscript for simplicity, θ_{null} can be defined as any value. This flexibility allows the testing framework to shift from merely evaluating whether an effect differs from zero to assessing whether it exceeds a substantively meaningful threshold.

Simulation

In our simulations we generate data under the global null (all $\theta = 0$) and we compute the frequency of $pd > c$, where $c = 0.975$. Operationally, this quantifies how often the procedure falsely excludes zero from the credible region (bounded by $(1 - c)$ and c), that is, incorrectly concludes that an effect exists (with a particular sign). We conduct a Monte Carlo simulation in which, for each condition, we generated families of size $m \in \{1, 2, 5, 10, 20\}$, with $n \in \{30, 50, 100\}$ observations per dataset, and repeated the procedure 10,000 times. In every

replication, data were generated under the global null: each dataset had true mean 0 and unit variance, $y \sim \mathcal{N}(0, 1)$. Each dataset was analyzed using a conjugate normal model with known variance $\sigma^2 = 1$ and prior $\theta \sim \mathcal{N}(0, \tau^2)$, where $\tau \in \{0.2, 0.5, 1, 1.5, 2\}$ spans skeptical to flat priors. We drew 10000 posterior samples of θ from the resulting normal posterior and computed *pd* as $\max(\Pr(\theta > 0 \mid y), \Pr(\theta < 0 \mid y))$ using these draws. For the frequentist comparison, we conducted two-sided one-sample z-tests.

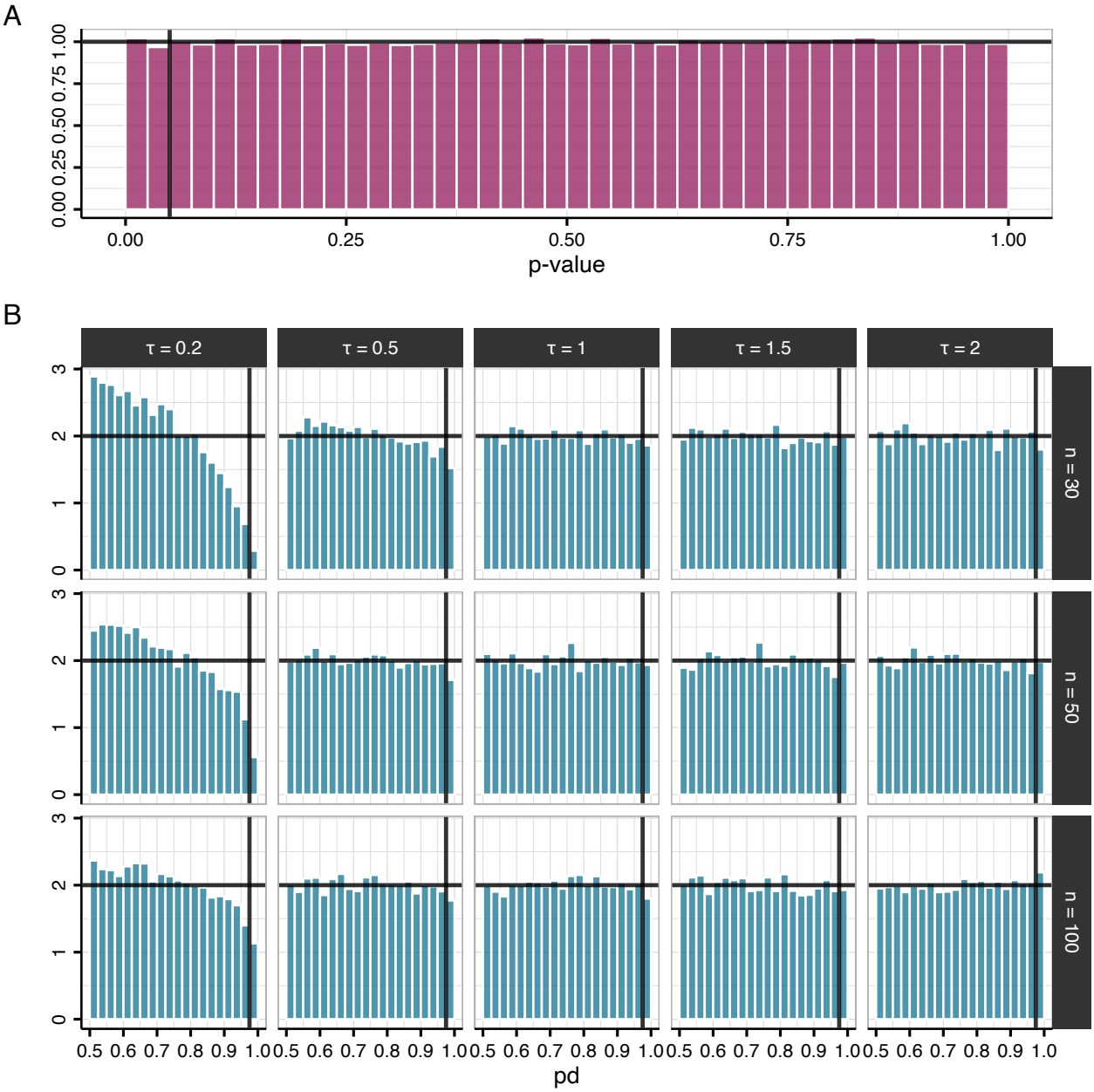
Prior influence

Under the null hypothesis, two-sided p-values are uniformly distributed on $[0, 1]$, and because *pd* can be mapped to the p-value, this implies a corresponding null reference behavior for *pd* on $[0.5, 1]$ in the continuous case. However, because the *pd* is a posterior probability about the sign of θ, its null distribution is not determined by the likelihood alone: it can shift depending on how strongly the prior concentrates probability near 0 (i.e., how readily the prior "expects" near-null effects). Showing this prior dependence at $m = 1$ makes the target of later multiplicity analyses explicit: even before any multiplicity is introduced, a fixed cutoff such as $pd > 0.975$ does not have a universal "null meaning" across priors and sample sizes.

As shown in Figure 3, *pd* exhibits behavior broadly analogous to a p-value when samples are large and the prior places relatively little mass near 0. However, increasing prior skepticism (i.e., concentrating more prior mass around 0) reduces the frequency of extreme *pd* values.

Figure 3

Null reference distributions for p-values and probability of direction. Panel A shows the histogram of two-sided one-sample z-test p-values from 10,000 replications under the global null ($\theta = 0$), with binwidth 0.025. Panel B shows the corresponding empirical distributions of the probability of direction (pd) computed from a conjugate normal posterior, stratified by sample size ($n = 30, 50, 100$; rows) and prior scale ($\theta \sim \mathcal{N}(0, \tau^2)$ with $\tau \in \{0.2, 0.5, 1, 1.5, 2\}$; columns). The vertical dashed line marks the commonly used evidential cutoff (pd = 0.975), which corresponds to a two-sided p-value of 0.05 under the standard conversion $p = 2(1 - pd)$.



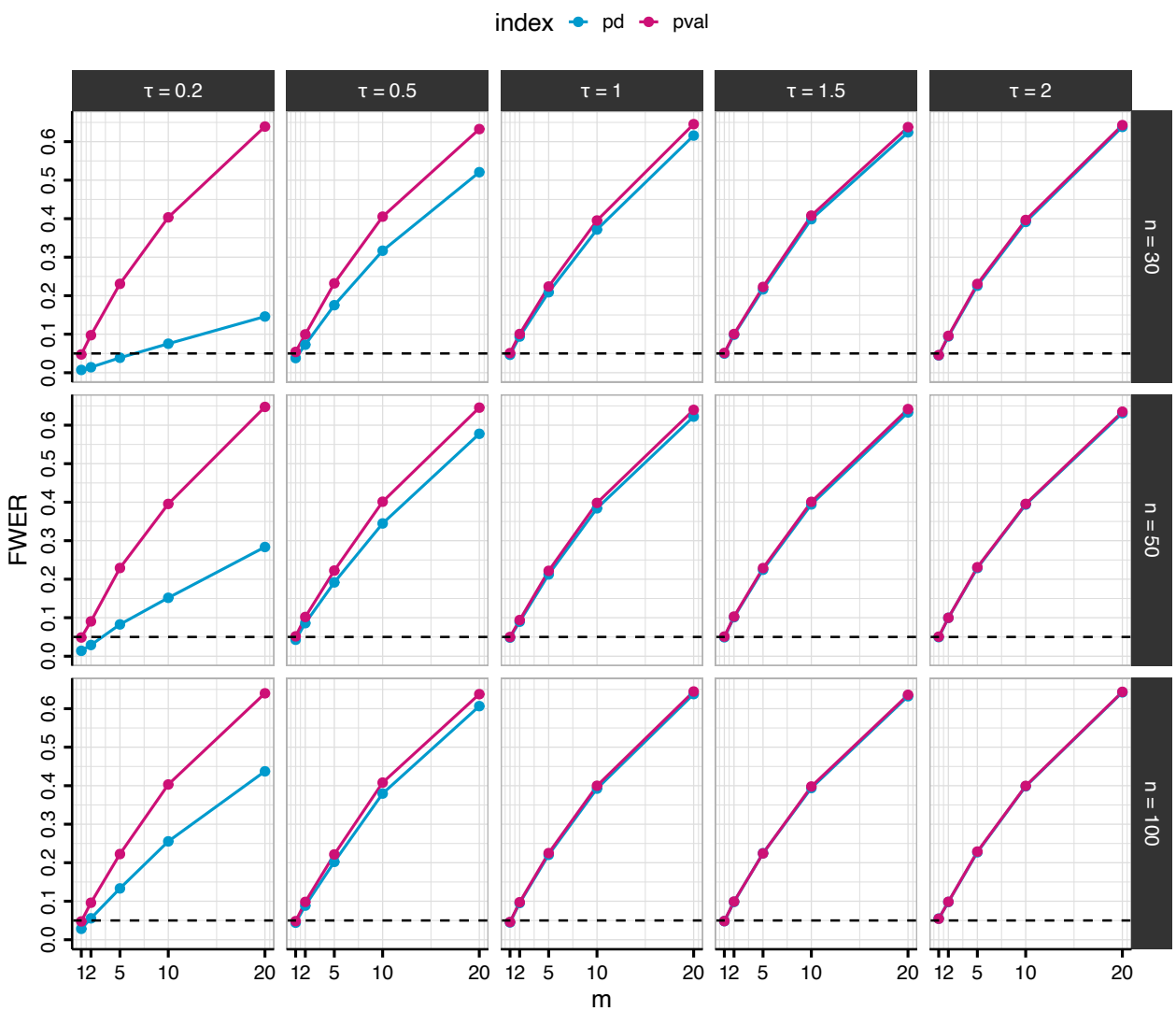
Behavior under multiplicity

Building on this baseline, multiplicity can be introduced as a second, distinct mechanism: even when each individual test has only a small probability of producing an extreme pd under H_0, the probability that at least one among $pd_1, \ldots, pd_m$ is extreme increases with the number of tests m. We quantify this familywise error rate (FWER) behavior by estimating the frequency of $pd > 0.975$ for families ranging from $m = 1$ to $m = 20$.

As expected, pd-based decisions behaves broadly analogously to their frequentist counterpart when sample sizes are large and the prior places relatively little mass near 0 (Figure 4). By contrast, as the prior becomes more skeptical (i.e., increasingly concentrated around 0), the FWER decreases.

Figure 4

Familywise error rates under the global null for p-values and probability of direction. The figure shows the estimated familywise error rate (FWER), defined as the probability of at least one false positive across m tests, as a function of the number of tests (m), based on 10,000 replications under the global null ($\theta = 0$ for all tests). The pink curve corresponds to declaring at least one discovery when at least one two-sided one-sample z-test p-value < 0.05, whereas the blue curve corresponds to declaring at least one discovery when at least one pd > 0.975 under the conjugate normal Bayesian model. Rows vary the per-test sample size ($n = 30, 50, 100$), and columns vary the prior scale ($\theta \sim \mathcal{N}(0, \tau^2)$ with $\tau \in \{0.2, 0.5, 1, 1.5, 2\}$).



The simulation results suggest that Type I error could, in principle, be reduced by adopting a more skeptical within-model prior for each parameter. However, using within-model prior skepticism as a general remedy for multiplicity is typically not the most appropriate response. This approach conflates two separate goals: (i) encoding substantive prior knowledge about a specific parameter value and (ii) calibrating a decision rule's error rate under repeated testing. When aiming to control the error rate, the prior should be chosen specifically for that purpose, but without compromising the accuracy of parameter estimation. For this reason, the approach developed below targets multiplicity at the level where it arises (the competition among hypotheses), by adjusting the prior odds assigned across competing hypotheses while leaving the within-model prior for any single parameter unchanged.

Probability of direction, posterior odds, and prior-odds adjustment

From posterior probabilities to posterior odds

To connect pd to posterior odds, consider the directional hypotheses $H_+ : \theta > 0$ and $H_- : \theta < 0$. These hypotheses partition the parameter space based on the sign of θ. If the prior for θ is continuous and symmetric about 0, then the two directions are equally plausible a priori: $\Pr(H_+) = \Pr(H_-) = 1/2$.

Without loss of generality, assume the posterior favors H_+, so that $pd = \Pr(H_+ \mid y) = \Pr(\theta > 0 \mid y)$, then the complement is given by $1 - pd = \Pr(\theta \leq 0 \mid y)$. The posterior odds comparing these two hypothesis are therefore:

$$\frac{pd}{1 - pd} = \frac{\Pr(H_+ \mid y)}{\Pr(H_{-0} \mid y)}.$$

For a continuous posterior distribution, the probability of any single point (e.g., $\Pr(\theta = 0 \mid y)$) is 0, which means that one could write $\frac{pd}{1-pd} = \frac{\Pr(H_+|y)}{\Pr(H_-|y)}$. However, we retain the notation H_{-0} in the denominator to explicitly indicate that the hypothesis includes the point null value of zero.

Applying Bayes' rule, these posterior odds can be decomposed into a Bayes factor and prior odds:

$$\frac{\Pr(H_+ \mid y)}{\Pr(H_{-0} \mid y)} = \mathrm{BF}_{+/-0} \times \frac{\Pr(H_+)}{\Pr(H_{-0})},$$

where BF+/−0 quantifies the relative likelihood of the data under $H+$ versus H_{-0}. Under prior symmetry, $\frac{\Pr(H_+)}{\Pr(H_{-0})} \approx 1$, which yields $\frac{pd}{1-pd} = \mathrm{BF}_{+/-0}$ (Casella & Berger, 1987; Marsman & Wagenmakers, 2017). Thus, the odds $\frac{pd}{(1-pd)}$ computed from posterior draws can be interpreted directly as evidence favoring $\theta > 0$ over $\theta \leq 0$ (or $\theta < 0$ over $\theta \geq 0$).

Multiplicity adjustment via post-hoc prior-odds

Assessing a single directional claim under a symmetric prior centered at the null assumes prior indifference. However, when m hypotheses are tested simultaneously, assigning prior probability $\frac{1}{2}$ to each directional claim implies a joint prior that rapidly concentrates probability on the event that at least one effect departs from 0. For instance, with $m = 10$ the implied prior probability that all effects are not in the favored direction (i.e., in the opposite direction with zero included) is $0.5^{10} \approx 0.001$.

In this scenario, a principled way to address multiplicity is to decrease the prior odds assigned to any particular directional claim as a function of m, while leaving the the Bayes factor component unchanged. Because the analysis is symmetric in the sign of θ and pd is defined as the posterior probability of the dominant direction, the same adjustment can be applied regardless of whether the posterior favors $\theta > 0$ or $\theta < 0$.

Concretely, let $\Pr(H_{1_i})$ and $\Pr(H_{0_i})$, where $i = 1, \ldots, m$, denote the multiplicity-adjusted prior probabilities assigned to the favored direction and its complement (i.e., opposite direction with zero included), respectively (with $\Pr(H_{0_i}) + \Pr(H_{1_i}) = 1$). The adjusted posterior odds are then:

$$\frac{pd_{\mathrm{adj}_i}}{1 - pd_{\mathrm{adj}_i}} = \frac{pd_i}{1 - pd_i} \times \frac{\Pr(H_{1_i})}{\Pr(H_{0_i})},$$

so that, equivalently,

$$pd_{\mathrm{adj}_i} = \frac{pd_i \cdot \Pr(H_{1_i})}{pd_i \cdot \Pr(H_{1_i}) + (1 - pd_i) \cdot \Pr(H_{0_i})}.$$

Thus, the multiplicity adjustment acts through the prior-odds factor $\Pr(H_{1_i})/\Pr(H_{0_i})$.

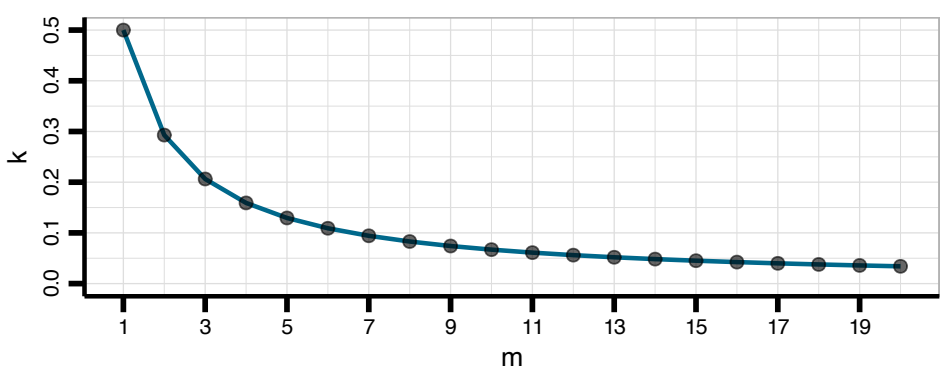
Choosing the prior-odds

Following the global-null formulation of Jeffreys (1938; see also Westfall et al., 1997), we introduce a global null event Q and its complement $\neg Q$, and assign equal prior probability $\Pr(Q) = \Pr(\neg Q) = 1/2$. Define, for each $i = 1, \ldots, m$, the "favored-direction" alternative $H_{1,i}$ as the directional hypothesis corresponding to the posterior-dominant sign (i.e., the direction used in $pd_i = \Pr(H_{1,i} \mid y)$), and let $H_{0,i}$ denote its complement (opposite direction with 0 included).

We then set $Q = \bigcap_{i=1}^{m} H_{0,i}$ (no parameter lies in its favored direction) and $\neg Q$ as its complement (at least one parameter lies in its favored direction). Under independence and a common per-hypothesis prior probability $k = \Pr(H_{1,i})$ for all j, this implies $\Pr(Q) = (1-k)^m$, so $(1-k)^m = 0.5$ and therefore $k = 1 - 0.5^{1/m}$. As m increases, the per-hypothesis prior probability $k = \Pr(H_{1,i})$ decreases, encoding increasing prior skepticism (Figure 5).

Figure 5

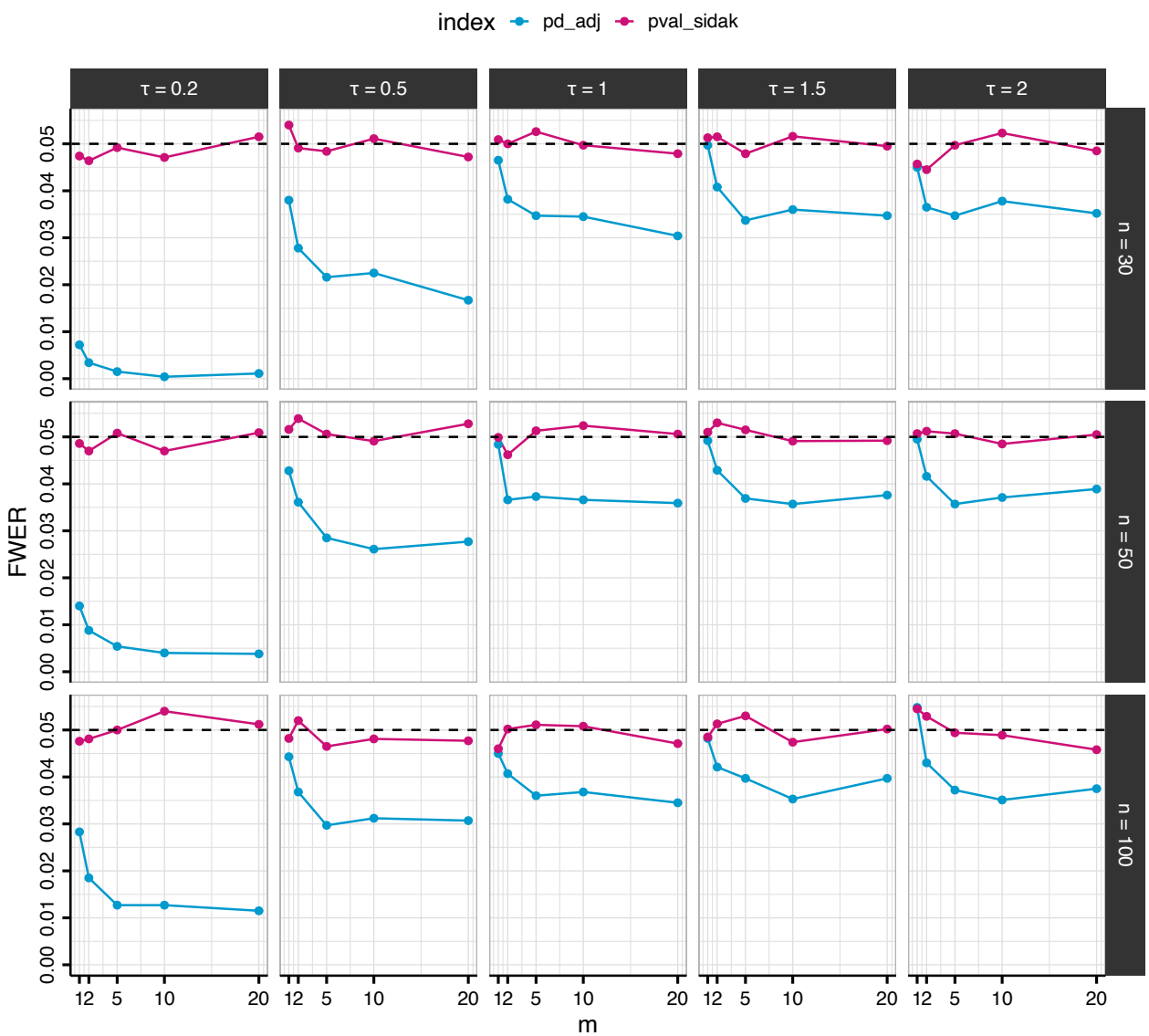
Prior probabilty k for each $H_{1,i}$ (where $i = 1, \ldots, m$) as a function of the total number of hypotheses m, assuming the joint null hypothesis has a probability of 0.5.



It is crucial to emphasize that the proposed Bayesian adjustment is valid only in the presented case, specifically when priors are symmetric about the null value. In the simulation study, we compare this method with the Šidák p-value adjustment ($p_{adj} = 1 - (1-p)^m$), which gives exact frequentist family-wise error rate control under independence. Although grounded in Bayesian prior-odds logic, the procedure exhibits behavior similar to frequentist multiplicity correction. However, it tends to be more conservative, even when both n and τ are large (Figure 6).

Figure 6

Familywise error rates under the global null for adjusted p-values and probability of direction. The figure shows the estimated familywise error rate (FWER), defined as the probability of at least one false positive across m tests, as a function of the number of tests (m), based on 10,000 replications under the global null ($\theta = 0$ for all tests). The red curve corresponds to declaring at least one discovery when at least one two-sided one-sample z-test p-value < 0.05, whereas the blue curve corresponds to declaring at least one discovery when at least one pd > 0.975 under the conjugate normal Bayesian model. Rows vary the per-test sample size ($n = 30, 50, 100$), and columns vary the prior scale ($\theta \sim \mathcal{N}(0, \tau^2)$ with $\tau \in \{0.2, 0.5, 1, 1.5, 2\}$).



Having linked the multiplicity adjustment to the global null event Q, the same construction yields a more general calibration directly in terms of $\Pr(Q)$. Specifically, instead of fixing $\Pr(Q) = 0.5$, researchers may encode background knowledge by choosing $\Pr(Q) = q$. As stated above, under a priori independence and a common per-hypothesis prior probability $k = \Pr(H_{1,i})$ for the favored-direction alternative, $\Pr(Q) = (1-k)^m$, so choosing q implies

$$(1-k)^m = q \quad \Rightarrow \quad k = 1 - q^{1/m} = \Pr(H_{1,i}).$$

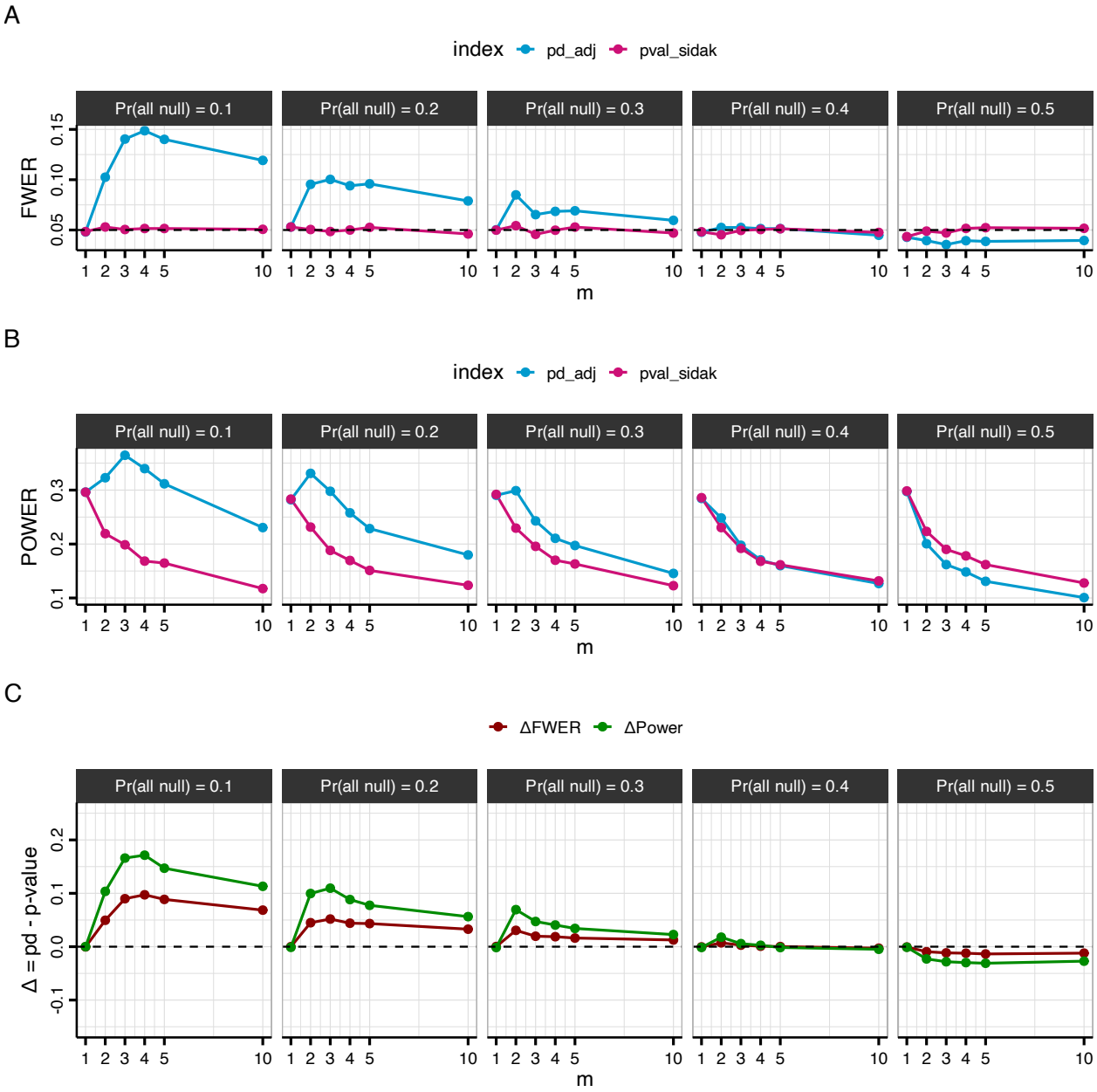
Prior knowledge guides q: when few (if any) non-null effects are expected, q should be large; when effects are well established, q can be smaller. For small m and sufficiently small q, however, the calibration can become anti-conservative: $\Pr(H_{0,i}) = 1 - k = q^{1/m}$ may fall below $1/2$, which implies $\Pr(H_{1,i}) > 1/2$ and therefore increases rather than decreases the odds for $H_{1,i}$. Because the goal of the adjustment is to guard against inflated false positives, it should never move posterior probabilities in this direction; accordingly, a simple safeguard is to leave posterior probabilities unchanged whenever $\Pr(H_{0,i}) < 1/2$ (equivalently, whenever $q^{1/m} < 1/2$).

Once q is fixed, the prior odds assigned to each individual hypothesis become a deterministic function of the family size m: larger families imply smaller k, and consequently a higher prior probability on the null for each test.

To illustrate the impact of q on both FWER and power, we consider two data-generating regimes: a global-null scenario in which $\theta_1 = \cdots = \theta_m = 0$, and an alternative scenario in which all but one effect are null ($\theta_1 = \cdots = \theta_{m-1} = 0$, $\theta_m = 0.2$). We then repeat the previous simulation procedure with $\tau = 2$ and $n = 50$, i.e. a setting in which *pd* and *p*-value behavior is broadly comparable, while varying only the number of tests $m \in \{1, 2, 3, 4, 5, 10\}$ and the prior parameter $q \in \{0.50, 0.40, 0.30, 0.20, 0.10\}$.

Figure 7

Familywise error rate (Panel A), power (Panel B), and difference between method (Panel C) for adjusted p*-values and probability of direction as a function of the number of tests (m) and prior* Pr(*all* $\theta = 0$). *The Šidák-adjusted* p*-value curve (pink) declares at least one discovery when any two-sided one-sample* z*-test satisfies* $p < 0.05$*, whereas the adjusted* pd *curve (blue) declares at least one discovery when any* $pd > 0.975$ *under the conjugate normal Bayesian model. Panel C shows the difference (adjusted pd minus Šidák-adjusted p-value) for FWER (under the global null) and power (under the global alternative).*



Across all values of m, Šidák-adjusted p-values maintain the family-wise error rate (FWER; Figure 7, Panel A) near the nominal level, but this control comes at the expense of diminished power (Figure 7, Panel B). By contrast, the adjusted *pd* procedure exhibits FWER and power trajectories that depend on the prior parameter q. For smaller values of q ($q \in \{0.30, 0.20, 0.10\}$), the tradeoff is favorable: the increase in power is larger than the corresponding increase in FWER, implying gains relative to the loss of error control (Figure 7, Panel C). Notably, $q = 0.4$ yields FWER and power curves that closely mirror the Šidák adjustment across the full range of m, suggesting that this calibration allows the Bayesian adjustment to approximate classical strong FWER control (Figure 7).

Application

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Discussion

Although *pd* is a descriptive posterior index, thresholding it into a decision rule and applying that rule repeatedly across many parameters induces a multiplicity problem, as the probability of at least one false claim increases with the number of decisions. The multiplicity adjustment proposed here targets this decision-level error while allowing researcher-specific knowledge to enter transparently through the prior-odds specification.

An advantage of this procedure is its applicability to *pd* values computed from posterior distributions obtained under different models. While the most coherent approach analyzes all relevant data within a single joint model that explicitly represents dependence (Gelman et al., 2012, 2013; Kruschke, 2011), this is not always feasible in practice, and researchers often report *pd* across multiple analyses within the same manuscript [**REFs**]. The procedure outlined here is specifically designed for these scenarios. It can also be easily generalized to cases with non-zero null values. This generalization is achieved by defining directional hypotheses relative to any chosen value, provided that the within-model priors for the tested parameters are set symmetrically around this value.

The procedure does assume independence between hypotheses when allocating global null

probability across m tests, treating each as providing independent evidence. When hypotheses are correlated through shared parameters, overlapping data structures, or measurement dependencies, independence-based adjustments overestimate the multiplicity penalty, yielding overly conservative estimates (Pesarin & Salmaso, 2010; Westfall & Young, 1993). A practical solution could be to replace the nominal number of tests m with an effective number m_{eff} that reflects the dependence structure. This can be achieved by computing a correlation matrix R that captures correlation among tests (e.g., posterior distributions from the same model) and mapping R to m_{eff} via eigenvalue-based methods (Cheverud, 2001; Del Giudice, 2021; Li & Ji, 2005). The resulting m_{eff} will generally depend on how R is defined and estimated, so it should be treated as a sensitivity parameter rather than a uniquely determined quantity.

Another important caveat is that this framework does not, by itself, deliver evidence for a point null (in the sense of assigning posterior support to $\theta = 0$); instead, it evaluates directional claims defined on inequalities (e.g., $\theta > \theta_{null}$ vs. $\theta \leq \theta_{null}$), where the complement includes the null value by construction. This contrasts with other Bayes factor approaches that represent the point null as a distinct model component by assigning it positive prior mass, thereby allowing direct statements of evidence in favor of the point-null hypothesis (Heck et al., 2023; Ly et al., 2016; Morey & Rouder, 2011).

For applied reporting, a two-tier strategy is recommended: (i) report the unadjusted *pd* for each parameter to preserve transparency, and (ii) report at least one multiplicity-adjusted measure based on the proposed prior-odds correction. When *pd* values are computed from posterior distributions obtained from different models, a conservative default is to report the adjustment induced by $\Pr(Q) = 0.4$, a skeptical but defensible choice that yields corrections analogous to the Šidák adjustment for p-values. When substantive knowledge motivates alternative priors, report both the conservative and the informed adjustment to demonstrate robustness to plausible prior assumptions.

Despite its limitations, our method provides a coherent, practical way to recalibrate *pd*-based decisions for multiple comparisons, making reported claims less likely to reflect chance

patterns across many parameters.

References

Albers, C. J., Kiers, H. A., & Ravenzwaaij, D. van. (2018). Credible confidence: A pragmatic view on the frequentist vs Bayesian debate. *Collabra: Psychology*, *4*(1).

Altoè, G., Bertoldo, G., Zandonella Callegher, C., Toffalini, E., Calcagnì, A., Finos, L., & Pastore, M. (2020). Enhancing statistical inference in psychological research via prospective and retrospective design analysis. *Frontiers in Psychology*, *Volume 10 - 2019*. https://doi.org/10.3389/fpsyg.2019.02893

Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society. Series B (Methodological)*, *57*(1), 289–300. https://doi.org/10.2307/2346101

Berger, J. O., & Sellke, T. (1987). Testing a point null hypothesis: The irreconcilability of p values and evidence. *Journal of the American Statistical Association*, *82*(397), 112–122.

Bernardo, J. M., & Smith, A. F. M. (1994). *Bayesian Theory* (1st ed.). Wiley. https://doi.org/10.1002/9780470316870

Berry, D. A., & Hochberg, Y. (1999). Bayesian perspectives on multiple comparisons. *Journal of Statistical Planning and Inference*, *82*(1), 215–227. https://doi.org/10.1016/S0378-3758(99)00044-0

Box, G. E. P., & Tiao, G. C. (1992). *Bayesian inference in statistical analysis* (1st ed.). Wiley. https://doi.org/10.1002/9781118033197

Casella, G., & Berger, R. L. (1987). Reconciling bayesian and frequentist evidence in the one-sided testing problem. *Journal of the American Statistical Association*, *82*(397), 106–111. https://doi.org/10.1080/01621459.1987.10478396

Chen, G., Taylor, P. A., Cox, R. W., & Pessoa, L. (2020). Fighting or embracing multiplicity in neuroimaging? Neighborhood leverage versus global calibration. *NeuroImage*, *206*, 116320. https://doi.org/10.1016/j.neuroimage.2019.116320

Cheverud, J. M. (2001). A simple correction for multiple comparisons in interval mapping genome scans. *Heredity*, *87*(1), 52–58.

https://doi.org/https://doi.org/10.1046/j.1365-2540.2001.00901.x

Cohen, J. (1994). The earth is round (p < .05). *American Psychologist*, *49*(12), 997–1003. https://doi.org/10.1037/0003-066X.49.12.997

Dawid, A. P. (1994). Selection paradoxes of bayesian inference. *Lecture Notes-Monograph Series*, *24*, 211–220.

Del Giudice, M. (2021). Effective dimensionality: A tutorial. *Multivariate Behavioral Research*, *56*(3), 527–542.

Edwards, W., Lindman, H., & Savage, L. J. (1963). Bayesian statistical inference for psychological research. *Psychological Review*, *70*(3), 193.

Erčulj, F., & Štrumbelj, E. (2015). Basketball shot types and shot success in different levels of competitive basketball. *PLOS ONE*, *10*(6), 1–14. https://doi.org/10.1371/journal.pone.0128885

Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B. (2013). *Bayesian data analysis* (3rd ed.). Chapman; Hall/CRC. https://doi.org/10.1201/b16018

Gelman, A., Hill, J., & Yajima, M. (2012). Why we (usually) don't have to worry about multiple comparisons. *Journal of Research on Educational Effectiveness*, *5*(2), 189–211. https://doi.org/10.1080/19345747.2011.618213

Gelman, A., & Tuerlinckx, F. (2000). Type s error rates for classical and bayesian single and multiple comparison procedures. *Computational Statistics*, *15*(3), 373–390. https://doi.org/10.1007/s001800000040

Gigerenzer, G. (2004). Mindless statistics. *The Journal of Socio-Economics*, *33*(5), 587–606. https://doi.org/https://doi.org/10.1016/j.socec.2004.09.033

Goeman, J. J., Solari, A., & Stijnen, T. (2010). Three-sided hypothesis testing: Simultaneous testing of superiority, equivalence and inferiority. *Statistics in Medicine*, *29*(20), 2117–2125. https://doi.org/https://doi.org/10.1002/sim.4002

Heck, D. W., Boehm, U., Böing-Messing, F., Bürkner, P.-C., Derks, K., Dienes, Z., Fu, Q., Gu, X., Karimova, D., Kiers, H. A. L., Klugkist, I., Kuiper, R. M., Lee, M. D., Leenders, R., Leplaa,

H. J., Linde, M., Ly, A., Meijerink-Bosman, M., Moerbeek, M., … Hoijtink, H. (2023). A review of applications of the Bayes factor in psychological research. *Psychological Methods*, *28*(3), 558–579. https://doi.org/10.1037/met0000454

Hogeveen, J., Mullins, T. S., Romero, J. D., Eversole, E., Rogge-Obando, K., Mayer, A. R., & Costa, V. D. (2022). The neurocomputational bases of explore-exploit decision-making. *Neuron*, *110*(11), 1869–1879.e5. https://doi.org/10.1016/j.neuron.2022.03.014

Hoijtink, H., Kooten, P. van, & Hulsker, K. (2016). Why bayesian psychologists should change the way they use the bayes factor. *Multivariate Behavioral Research*, *51*(1), 2–10.

Jeffreys, H. (1938). Significance tests when several degrees of freedom arise simultaneously. *Proceedings of the Royal Society of London. Series A. Mathematical and Physical Sciences*, *165*(921), 161–198. https://doi.org/10.1098/rspa.1938.0052

Jevremov, T., & Pajić, D. (2024). Bayesian method in psychology: A bibliometric analysis. *Current Psychology*, *43*(10), 8644–8654.

Kelter, R. (2023). How to choose between different bayesian posterior indices for hypothesis testing in practice. *Multivariate Behavioral Research*, *58*(1), 160–188.

Kruschke, J. K. (2011). Bayesian assessment of null values via parameter estimation and model comparison. *Perspectives on Psychological Science*, *6*(3), 299–312. https://doi.org/10.1177/1745691611406925

Kruschke, J. K. (2018). Rejecting or Accepting Parameter Values in Bayesian Estimation. *Advances in Methods and Practices in Psychological Science*, *1*(2), 270–280. https://doi.org/10.1177/2515245918771304

Li, J., & Ji, L. (2005). Adjusting multiple testing in multilocus analyses using the eigenvalues of a correlation matrix. *Heredity*, *95*(3), 221–227.

Limbachia, C., Morrow, K., Khibovska, A., Meyer, C., Padmala, S., & Pessoa, L. (2021). Controllability over stressor decreases responses in key threat-related brain areas. *Communications Biology*, *4*(1), 42. https://doi.org/10.1038/s42003-020-01537-5

Lindley, D. V. (1972). Bayesian statistics a review. In *Bayesian statistics* (pp. 1–74). Society for

Industrial; Applied Mathematics. https://doi.org/10.1137/1.9781611970654.ch1

Ly, A., Verhagen, J., & Wagenmakers, E.-J. (2016). Harold jeffreys's default bayes factor hypothesis tests: Explanation, extension, and application in psychology. *Journal of Mathematical Psychology*, *72*, 19–32. https://doi.org/https://doi.org/10.1016/j.jmp.2015.06.004

Makowski, D., Ben-Shachar, M. S., Chen, S. H. A., & Lüdecke, D. (2019). Indices of effect existence and significance in the bayesian framework. *Frontiers in Psychology*, *Volume 10 - 2019*. https://doi.org/10.3389/fpsyg.2019.02767

Marsman, M., & Wagenmakers, E.-J. (2017). Three insights from a bayesian interpretation of the one-sided p value. *Educational and Psychological Measurement*, *77*(3), 529–539. https://doi.org/10.1177/0013164416669201

Mathar, D., Erfanian Abdoust, M., Marrenbach, T., Tuzsus, D., & Peters, J. (2022). The catecholamine precursor tyrosine reduces autonomic arousal and decreases decision thresholds in reinforcement learning and temporal discounting. *PLOS Computational Biology*, *18*(12). https://doi.org/10.1371/journal.pcbi.1010785

Morey, R. D., & Rouder, J. N. (2011). Bayes factor approaches for testing interval null hypotheses. *Psychological Methods*, *16*(4), 406–419. https://doi.org/10.1037/a0024377

Pesarin, F., & Salmaso, L. (2010). The permutation testing approach: A review. *Statistica*, *70*(4), 481–509. https://doi.org/10.6092/issn.1973-2201/3599

Pfadt, J. M., Bartoš, F., Godmann, H. R., Waaijers, M., Groot, L., Heo, I., et al. (2025, October 30). *A methodological metamorphosis: The rapid rise of bayesian inference and open science practices in psychology*. OSF Preprints. https://doi.org/10.31234/osf.io/ck3js_v1

Pruzek, R. M. (2016). *An introduction to bayesian inference and its applications* (pp. 305–334). Routledge.

Rozeboom, W. W. (1960). The fallacy of the null-hypothesis significance test. *Psychological Bulletin*, *57*(5), 416–428. https://doi.org/10.1037/h0042040

Schnell, P. M., Fiecas, M., & Carlin, B. P. (2020). **credsubs** : Multiplicity-adjusted subset

identification. *Journal of Statistical Software*, *94*(7). https://doi.org/10.18637/jss.v094.i07

Scott, J. G., & Berger, J. O. (2006). An exploration of aspects of bayesian multiple testing. *Journal of Statistical Planning and Inference*, *136*(7), 2144–2162.

Scott, J. G., & Berger, J. O. (2010). Bayes and empirical-Bayes multiplicity adjustment in the variable-selection problem. *The Annals of Statistics*, *38*(5), 2587–2619. https://doi.org/10.1214/10-AOS792

Smid, S. C., & Winter, S. D. (2020). Dangers of the defaults: A tutorial on the impact of default priors when using bayesian SEM with small samples. *Frontiers in Psychology*, *Volume 11 - 2020*. https://doi.org/10.3389/fpsyg.2020.611963

Tendeiro, J. N., & Kiers, H. A. (2019). A review of issues about null hypothesis bayesian testing. *Psychological Methods*, *24*(6), 774.

Tendeiro, J. N., Kiers, H. A., Hoekstra, R., Wong, T. K., & Morey, R. D. (2024). Diagnosing the misuse of the bayes factor in applied research. *Advances in Methods and Practices in Psychological Science*, *7*(1), 25152459231213371.

Van De Schoot, R., Winter, S. D., Ryan, O., Zondervan-Zwijnenburg, M., & Depaoli, S. (2017). A systematic review of bayesian articles in psychology: The last 25 years. *Psychological Methods*, *22*(2), 217.

Westfall, P. H., Johnson, W. O., & Utts, J. M. (1997). A bayesian perspective on the bonferroni adjustment. *Biometrika*, *84*(2), 419–427.

Westfall, P. H., & Young, S. S. (1993). *Resampling-based multiple testing: Examples and methods for p-value adjustment*. Wiley.

Zwet, E. van, & Cator, E. A. (2021). The significance filter, the winner's curse and the need to shrink. *Statistica Neerlandica*, *75*(4), 437–452. https://doi.org/10.1111/stan.12241

Zwet, E. van, & Gelman, A. (2022). A proposal for informative default priors scaled by the standard error of estimates. *The American Statistician*, *76*(1), 1–9. https://doi.org/10.1080/00031305.2021.1938225